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RevisitingAlmaAta:ABlueprintforCancer Care

Javier-DavidBenitez-Fuentes,MD,MSc[1,2],
RodrigoLastradelPrado,MD,MSc[2,3],
MiguelBorregon-Rivilla,MD,MSc[1],
AliciadeLunaAguilar,MD,MSc[2,4],
Antonio-DavidLazaro-Sanchez,MD,MSc,PhD[2,4],
AsiaFerr´andez-Arias,MD[1],PaulaRodr´ıguezPay´a,MD[1],
BeatrizGrauMirete,MD[1],TeresaQuintanarVerduguez,MD[1],
ElenaAsensioMartinez,MD[1],PatriciaIranzo,MD,PhD[2,5],
AnaCallejo,MD,PhD[2,6],MaraCruellasLapeña,MD,PhD[2,3],
JacoboGómezUlla,MD,MSc[2,7],and
AlvaroRodriguez-Lescure,MD,PhD[1]

Abstract
Despitesignificantadvancesinoncology,cancercaregloballycontinuestofacecriticalchallenges,includingstarkdisparitiesin access, insufficient preventive focus, fragmented primary health care (PHC) integration, unsustainable financing models, workforce shortages,andinadequatecommunityinvolvement.ThispaperrevisitstheAlmaAtaDeclaration'sprinciples—healthequity, universal access, preventive care, and community participation—as a conceptual framework to address these persistent issues in cancer care. We highlight opportunities to strategically integrate oncology services within strengthened PHC systems, balancing centralizedspecialistresourceswithdecentralizedcommunity-basedcare.Evidencefromdiversesettingsillustrateshowre-inforcing PHC infrastructures enhances preventive measures, early detection, and survivorship care, thus mitigating geographic and socioeconomicdisparities.Sustainablefinancingmechanismsandtargetedworkforcestrategies,includingtask-shiftingand multidisciplinary training, are proposed as essential components. Effective community engagement models demonstrate improved care relevance, acceptance, and outcomes. Additionally, we emphasize the critical role of health policy alignment with universal **health coverage objectives, roubst pharmacoeconomic evaluations, and evidence-based national cancer control plans. Integrating**
AlmaAta'sprinciplesintocontemporaryoncologyprovidesaviable,scalablemodeltoadvanceequitable,accessible,andsus-tainable cancer care globally, laying the theoretical groundwork for future research initiatives and informed policy development.

PlainLanguageSummary
Why was this study done? Despite many medical advances, cancer care remains very unequal globally. Many people worldwide stillhavepooraccesstocancerservicesbecauseofhightreatmentcosts,lackoftrainedmedicalstaff,unevenhealthcare

[1]DepartmentofMedicalOncology,FISABIO,HospitalGeneralUniversitariodeElche,Alicante,Spain
[2]RIOWorkingGroup,Zaragoza,Spain
[3]DepartmentofMedicalOncology,HospitalClinicoUniversitarioLozanoBlesa,Zaragoza,Spain
[4]DepartmentofMedicalOncology,HospitalGeneralUniversitarioMoralesMeseguer,Murcia,Spain
[5]DepartmentofMedicalOncology,HospitalUniversitariValld'Hebron&Valld'HebronInstituteofOncology(VHIO),Barcelona,Spain
[6]DepartmentofMedicalOncology,HospitalUniversitariodeBurgos,Burgos,Spain
[7]DepartmentofMedicalOncology,HospitalUniversitariodeBadajoz,Badajoz,Spain

CorrespondingAuthor:
Javier David Benitez Fuentes, Department of Medical Oncology, FISABIO, Hospital General Universitario de Elche, Alicante, Carrer Almazara 11, Elche 03203, Spain.
Email:

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resources,andweakhealthcaresystems.ThisworkexploreshowtheprimaryhealthcareprinciplessetbytheAlmaAta Declaration—a major global health agreement from 1978 aimed at ensuring accessible health care for all—could help improve cancer care today. What did the researchers do? We reviewed challenges faced by current cancer care systems, including poor prevention,unequalaccesstocare,hightreatmentcosts,shortagesofspecialisthealthcareworkers,andlimitedcommunity involvement. We then proposed a way to improve these issues by applying the Alma Ata principles—such as universal access, prevention-focusedhealthcare,andcommunityinvolvement—tocancercare.Wealsohighlightedsuccessfulreal-world examples where these ideas have improved cancer services in different countries. What did the researchers find? Strengthening primaryhealthcarecouldsignificantlyimprovecancer outcomesbyhelpingtodetectcancersearlier,preventingmanycases, reducing healthcare costs, and making care fairer for everyone, no matter their location or income. Involving local communities directlyinhealthcareplanningandcreatingabalancedsystemcombiningspecializedcentersandlocalhealthservicescould benefitpatientsgreatly.WhatdothefindingsmeanPlayingtheAlmaAtaprimaryhealthcareapproachtocancercarecan make it more accessible, affordable, and effective worldwide. This provides a clear roadmap for researchers, healthcare leaders, andpolicymakersseekingfairerandbettercancercareforallpeople.

Keywords
AlmaAta,primaryhealthcare,disparities, health equity,health policy,global oncology,cancer, health systems,publichealth, universalhealthcoverage

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Introduction

TheAlmaAtaConferenceDeclaration(AACD),issuedin 1978duringtheInternationalConferenceonPrimaryHealth Careco-hostedbyWHOandUNICEF,revolutionizedglobal health care by establishing primary health care (PHC) as essential and health as a fundamental human right.[1]Prior to this landmark conference, access to essential health care services was severely limited,especiallyinruralregionsandamongimpoverished urbanpopulations.Healthsystemswereprimarilyreactive, disease-oriented,andheavilycentralizedinurbanhospitals, whichperpetuatedsignificanthealthdisparities.[2]TheAACD introducedatransformativevisionemphasizingPHCasacornerstoneofhealthsystems,advocatinghealthequity,universal accessibility,communityparticipation,andaholisticapproach thatexplicitlyaddressessocialdeterminantsofhealth.[2]

Followingthedeclaration,globalhealthstrategiesincreasinglyembracedcommunity-orientedapproaches,focusingon preventivecareandintersectoralcollaboration.InLowand Middle-IncomeCountries(LMICs),thesestrategiessubstantiallyimprovedhealthcareaccessdespitepersistenteconomic and logistical challenges.[3]The AACD's call for universal health care(UHC)catalyzednationalandinternationaleffortstoreducehealthdisparitiesandenhanceaccesstohealthcarefor underservedpopulations.However,fullyrealizingtheAlma Ata goals has been hindered by economic constraints, emerging healthcrises,andgeopoliticalchanges.[4]In2018,theAstana DeclarationmarkedthE40thanniversaryofAlmaAta,reaffirmingtheenduringimportanceofPHCandrenewingthe globalcommitmenttoUHC,emphasizingAACD'songoing relevancetocontemporaryglobalhealthpolicy.[4]

Within oncology, the core principles outlined by the AACD remainrelevant.Despiteremarkableadvancementsincancer

diagnosisandtherapeutics,contemporaryoncologyconfronts persistent challenges including significant disparities in access tocare,highcostsofinnovativetreatments,andanover-whelmingfocusonadvancedtherapiesofteninaccessibleto thosemostinneed.Theseissuesfrequentlyovershadowvital preventivestrategies.[5–7]Additionally,thecentralizationof oncology services within specialized urban hospitals continues to widen gaps in both access and quality, particularly affecting availability of novel therapies and clinical trial participation.[8–10] Thismanuscript,structuredasanarrativereviewandpolicy analysis,exploreshowintegratingAACD'sprinciplesintoon-cologycanhelpreshapecancercaresystems,advancingequity, strengtheningpreventiveapproaches,andenhancingPHCand communityparticipation.Weproposeaconceptualframework (Figure 1) that aims to align the foundational AACD's principles—healthequity,universalaccessibility,preventivecare,andcom-munityparticipation—withcontemporaryoncologypractice.

CurrentChallengesofOncologyCare

Cancercontinuestobeasignificantpublichealthchallenge worldwide. As of 2022, the global cancer burden escalated to anestimated20.3millionnewcasesandnearly10.9million cancer-relateddeaths,projectedtoreach35millioncasesby 2050duetoagingpopulations,lifestylechanges,andin-creasedriskfactorprevalence.[6]Theresponsibilityforthe treatmentofthesepatientsfallsinthehandsofmodernon-cology,afieldthathasmaderemarkablestrideswiththe advent of precision medicine and novel therapies. The current landscape,thoughmarkedbytheseadvancements,now confrontsacomplexarrayofcriticalchallengesthatthreaten equitableaccess,sustainablefinancing,andtrulypatient-centeredoutcomes(Figure2).

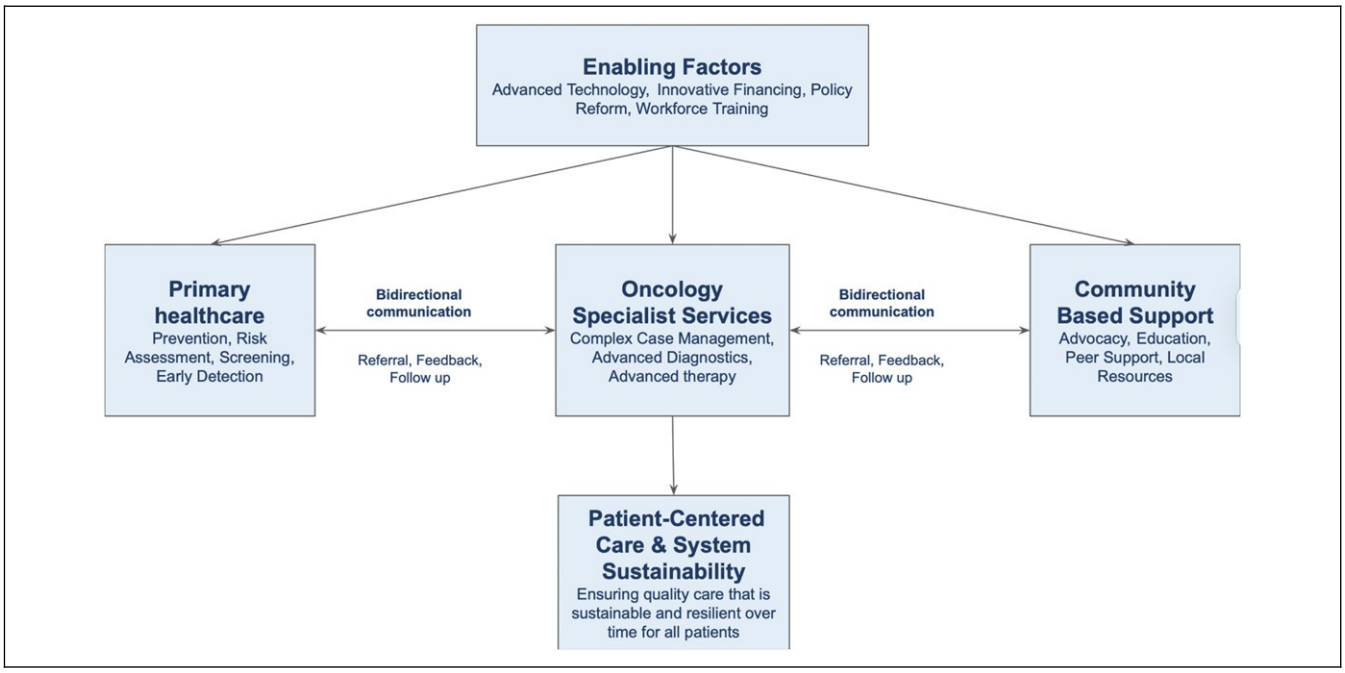

Figure 1.Conceptual Framework Integrating Alma-Ata Principles Into Oncology. The Model Links Primary Healthcare, Specialized Oncology Services, and Community-Based Support, Backed by Enabling Factors, to Deliver Equitable, Sustainable Patient-Centered Care Across the CancerContinuum

- CentralizationofOncologyCarogyserviceswithinacademicce hindersequitableaccesstoadva nologies,research,andclinicaltrials,particularlyaf-fectingpopulationsinremoteorsocioeconomically disadvantagedareas.[11–13]Conversely,decentralizing carecouldleadtosuboptimaltreatments,inefficient resourceuse,andalackofspecializedexpertise.[8,14] Thereisacriticalneedforresearchoneffectivelyin-ofexpenditurecomparedtotradi spending.[19]
- Global Cancer Care tegrating both centralizatior models. Existing studies on this are primarily conducted inhigh-incomecountries(HICs)andoftenfailtoad-dress population-specific needs at the country or health systemlevel.[8,10,14]

cancercareismarkedbysignificantdisparities,par-ticularlyinLMICs,wherelessthan25%ofpatients haveaccesstosafe,cost-effective,ortimelysurgical interventions.[5]Geographicandsocioeconomicfactors drasticallyimpactoutcomes,withovertwomillion womenannuallyfacingsurvivalrisksfrombreastor cervicalcancerduetolimitedaccess.[5,20–22]Inmany regions,theabsenceofUHCfurtherrestrictsaccessto cancercare,creatingaglobalscenariowherebothlo-cationandfinancialmeansdeterminetheleveland qualityofcare.[23–26]Evenwithinhigh-incomeregions

- Focusontreatmentratherthanprevention: likeEuropeandcountriesthemselves,c treatment-centricapproachmaycurenon-metastatic disease,itisusuallyonlymarginallyeffectiveinad-

vary between different geographic areas due to uneven resourceallocation.[27]

vancedstages.Thisfocusoftenoverlooksthe stantialbenefitsofearlyinterventionandpreventive measures,whichincludeprimarypreventionthrough health promotion and secondary prevention via targeted screening for specific tumors.[15–17]It is noteworthy that morethan50%ofcancercasescouldpotentiallybe

- CommunityParticipation:Communityen

prevented.[14]Investinginpreventionrequiresreallocating resources not only within oncology but across the entirehealthcaresystem.[14,18]Someestimatessugge st thatinvestinginpublichealthforpreventioncouldbe

cancer care remains inadequate in many settings, limiting theeffectivenessandculturalrelevanceofcancerprograms.Thisengagementoperatesintwodirections, shapinghealthstrategiesbyhighlightingcriticalneeds andperspectives,andassessingpatientsatisfactionand outcomes.Bothfacetsareincreasinglyvitalforresponsive,effectivecare.Engagingcommunitiesand enhancing public awareness are essential steps towards a more inclusive and effective approach to cancer care.[28,29]

- ChallengesinSustainableFinancing:Sustainablefinancing faces significant problems, particularly

significantly
up to four times

Figure2.CurrentChallengesofOncologyCare

regarding inequity in resource allocation and the costs of cancertreatments.[7]InLMICs,underfundedandover-stretched health systems hamper adequate care, while in HICs, rising therapy expenses lead to financial toxicity for patients, families, and the broader health system.[30,31] Advancedtherapiesoftencomeatprohibitiveprices, and the pharmaceutical industry's influence on escalating costsunderscorestheurgencyforapharmacoeconomic frameworkthatensuresevidence-based,consistentde-cisionsaboutwhichinterventionstofund.[7,30–32]Gov-ernments must also reflect on how heavily cancer therapy developmentreliesonthepharmaceuticalindustry,as inadequate support for large-scale academic projects can limittherationaluseofresources.[33–35]

oftenlackstransparent,evidence-basedpharmacoeco-nomic frameworks.[27]Collaboration with academia and clinicians in policy development is usually insufficient, leadingtohealthplansthatdonotfullymirrorpop-ulationneeds.Unresolvedtensionsbetweenprivatiza-tionandpublicinvestmentfurthercomplicatesystem resilience,equity,andviability.[27,46,47]

IntegratingAlmaAtaPrinciples intoOncology

IntegratingtheAACDprinciplesintooncologyrequiresa clearshifttowardequity,accessibility,andcommunity-orientedcare.Effectivecommunityengagement,robustre-ferral networks, and coordinated care

- ShortageofSpecialistWorkforce:Oncolog critical global shortage of specialists, notably in LMICs, severelyrestrictingeffectivecancermanagement.In regionsfromAfricaandAsia,oneoncologistmaybe responsibleforover1000newcancerpatients,re-flectingsignificantdisparitiesinworkforceavailabil-ity.[36]This deficit extends beyond physicians, including shortagesinskillednurses,technicians,anesthesiolo-

thisintegratedapproach.Thissectiondiscusseshowsuch integrationcanbepracticallyrealized,aligningwiththe conceptualframeworkpresentedearlier(Figure1),which explicitlylinksprimaryhealthcare,specializedoncology services,andcommunity-basedsupporttodeliverequitable, sustainable,andpatient-centeredcancercare.

gists, palliative care specialists, and PHC integraltomultidisciplinarycancercare.[5,3]

BalancingCentralizationandDece CancerCare

. GapsinPHC-

frastructure,hinderstimelycancerdetection,referral, and follow-up.[39,40]PHC is recognized as a fundamental elementofahealthsystem,yetinsufficientfunding, personnelandorganizationalsupporthaveledtosig-nificant gaps in care delivery. This shortfall, along with insufficientcommunicationchannelsamongPHC,on-cology, and supportive care teams, delays diagnosis and treatmentinitiation,ultimatelyaffectingpatientsatis-faction and outcomes.[29,41–43]This under-resourced and disjointedapproachleadstosuboptimalpatientout-

Cancercaresystemsmuststrategicallybalancecentralizing specializedservicesinhigh-volumecenterswithdecentral-
izingcareclosertocommunities.HICscountrieshaveoften pursuedcentralizationforcomplexcancersurgeriesand
treatmentsduetoevidencethathighercasevolumesmay improveoutcomes.Forexample,theNetherlandsintroduced
minimumvolumestandardsforpancreaticandesophageal cancersurgeries,significantlyenhancingsurvivalrates.[48]
However,centralizationincreasestraveldistances,creating accessbarriersthatdisproportionatelyaffectdisadvantaged
comesandincreasedburdenonhealthcares populations, as observed in the USA,
. geographicandsocioeconomicinequities.
GovernmentStrategiesandPolicyChalleng
policies frequently fail to match actual fiscal resultinginunrealisticobjectivesandineffectivere-source
capacities, allocation for cancer care.[44,45]Decision-making

SomeLMICs,traditionallyreliantoncentralizedservices inmajorcities,areevaluatingdecentralizationapproaches suchasthe“hub-and-spoke”model.Kenya,forinstance,

decentralizedoncologyservicesbyestablishingregional cancercenterswithinprovincialhospitals,significantlyen-hancing patient accessin underservedruralcommunities and facilitatingearlierdiagnosisandtreatment.[51]Similarinitia-tives in the Democratic Republic of Congo demonstrated that decentralizationcombinedwithlocalcapacity-buildingef-fectivelyreducesmorbidityandmortalitybyovercoming geographicbarriers.[52]InNorway,illustratingsuccessfulde-centralizationinanHIC,theOrkdalmodelintegratedon-cology and palliative care at local hospitals, improving timely access and patient outcomes through decentralized, community-levelservicedelivery.[53]Similarly,Italy'sTerri-

reforms under the Family Health Strategy expanded access to underservedpopulations,costingonlyabout$50perperson annually,reducinghospitalizationsandmortality.[64,65]

InHICs,patientnavigationprogramshavehelpedunder-servedgroupsovercomeobstaclesandreceivetimely cancerdiagnosesandtreatment,reducingoutcomegaps.[66] TheBostonPatientNavigationResearchProgram,effec-tively helped underserved groups overcome barriers to care. Inthisprogram,navigationwasassociatedwithasignifi-cantlyhigherlikelihoodofachievingdiagnosticresolu-tion.[67] A historical case from an underserved U.S. community(Harlem,NewYork)alsoshowedthatcom-bining free screening with patient navigation, as a andoftendeliveredthroughPHCandcommunityprograms. Tobacco control interventions integrated within PHC in India, usingcommunityhealthworkersforeducationandcessation support,couldreducesmokingrates,potentiallylowering tobacco-relatedcancerincidence.[59]InSouthAfrica,brief interventionsforalcoholreductionembeddedwithinPHC visitshaveeffectivelyreducedexcessivealcoholconsump-tion.[60]In Rwanda, nationwide HPV vaccination implemented through PHC and school-based platforms achieved over 90% coveragerate.[61]Thailand’sintegrationofhepatitisBvacci-nation into routine PHC immunization programs decreased the prevalence of hepatitis B infection among young adults from 8%-10%tolessthan1%,whichmayreducelivercancer incidenceovertwodecades.[62]

EquityinCancerCare

Achievingequityincancercareinvolvesensuringuniversal access irrespective of geographic or socioeconomic status. The WHOemphasizesthatstrongPHCiskeyforhealthsystems and equitable cancer outcomes.[63]For example, Brazil’s PHC

torialOncolog community hospitals,significantlyreducingpatienttravel burdenandachievinghighpatientsatisfactionbydelivering careclosertohome.[54]

Anoptimalapproachintegratesbothmodelsthrougha tieredsystem: high-complexity treatments remain centralized tomaintainquality,whilepreventivecare,earlydetection, basicchemotherapy,andsurvivorshipcarearedecentralized intoPHCorcommunityoncology,connectedbyrobustre-ferralmechanisms.[48,51,52]

EmphasisonPreventionandEarlyDetection

Preventivecareiscrucialinoncologytoreducecancerin-cidence and mortality.PHCteamsareuniquely positioned to leadcancerpreventionthroughlifestyleinterventions,vac-cination,andscreeningprograms.[55,56]

InHICs,preventiveinitiativesintegratedintoPHChave demonstratedeffectiveness.Forexample,structuredlifestyle interventions in the USA and the UK have reduced risk factors like obesity and tobacco use.[57,58]In LMICs, where resources for advanced care are limited, prevention is even more critical

community-based support intervention, dramatically closed thesurvivalgapforpoorpatients.Breastcancer5-year survivalinHarlemimprovedfrom39%to70%afterin-troducingthesemeasures.[68]InLMICs,equitycanbe promotedbybringingcloserservicestothecommunity. Rwanda'sruralButaroCancerCenter,forinstance,brings chemotherapy,surgery,andpalliativecaretoremoteareas andevenprovidestransportandlodgingforpatientswho otherwisewouldlackaccesstocare.[69]

CommunityInvolvementandParticipation

AACDaffirmedthatpeoplehaveboththerightanddutyto participateinplanningandimplementinghealthcare.Com-munityhealthworkers(CHWs)andvolunteersplaycrucial rolesinLMICsbyraisingawareness,facilitatingscreenings, andprovidingfollow-upcare.Asystematicreviewfound CHWsinLMICsplayanimportantroleincommunityedu-cation, outreach for screening, and follow-up.[70]For example, acommunity-integratedcervicalscreeninginitiativeinrural Uganda(theASPIREMayugetrial)increasedscreening participation by deploying CHWs for door-to-door education andself-samplingoutreach.[71]Anotherexamplecomesfrom Kerala,India,whereacommunity-drivenpalliativecare networkmobilizedlocalvolunteersalongsidehealthprofes-sionals to care for cancer patients at home. Over 15 000 trained volunteers,withsupportfromnursesanddoctors,nowhelp providehome-basedcareinalldistrictsofKerala,covering morethan60%ofpatientsinneed.[72,73]

HICsarealsoembracingcommunityinvolvement.For instance, community-based participatory research (CBPR) can engage local stakeholders in designing and evaluating cancer prevention and intervention programs, enhancing their cultural relevanceandeffectiveness.ArecentstudyutilizedCBPR approaches to adapt educational materials for cancer screening amongNativeAmericancommunities,resultinginculturally tailored resources.[74] Additionally, partnerships between healthcaresystemsandcommunityorganizationscansuc-cessfullytailorcancerawarenesscampaignsfordifferent populations.[75]Church–academicpartnershipswithAfrican Americancongregationshavesuccessfullydeliveredtailored cancerscreeningprogramstohelpreducedisparities.[76]

GlobalCancerCareDisparities

Globalcancercaredisparitiesremainprofound,withvastly differentsurvivaloutcomesbetweenHICsandLMICs,il-lustrated by breast cancer survival rates exceeding 89% in the United States but below 66% in India and other LMICs.[21,77,78] Similardisparitiesexistacrossothermalignancies,high-lighting a persistent gap in cancer care between high-resource settingsandLMICsordisadvantagedpopulationswithin wealthiernations.[79–81]

Centraltothesedisparitiesisunevenaccesstoessential cancerprevention,screening,anddiagnosticservices.For instance,over40%ofHICsreportmoderate-to-highcervical cancerscreeningcoverage,whilefewerthan4%oflow-incomecountriesachieveeven50%coverage.[82]Limitedin-frastructure,includingpathology,radiology,andspecialist surgical facilities, further exacerbates late-stage presentations, significantlyworseningoutcomes.[36,83]

StrengtheningPHCisessentialtomitigatingthesedis-parities.RegularengagementwithPHCpriortocancerdi-agnosissignificantlyreducesthelikelihoodofmetastatic disease and lowers cancer-specific mortality, as demonstrated intheU.S.VeteransAffairssystem.[84]ComplementingPHC, UHCensuresequitableaccesstoessentialserviceswithout financialhardship,makingitcrucialforsustainablecancer control,particularlyinresource-constrainedsettings.Inte-gratingcancerservicesintoPHCunderUHCframeworks facilitatesproactivemanagementthroughoutthecancer continuum,frompreventionandearlydetectiontosurvivor-shipandpalliativecare.[82,85]

ChallengesinSustainableFinancing

Cancercarefacesfinancialchallengesdrivenbyrapidlyes-calatingtreatmentcostsandrisingglobalcancerincidence. Between2020and2050,globalcancercareexpensesare projectedtoreachapproximately$25.2trillion.[86]Thiseco-nomicburdencreatesseverefinancialhardshipforbothpa-tientsandhealthcaresystems.Overhalfofcancerpatients globally (56.1%) experience catastrophic health expenditures, oftenresultinginseveredebt,assetloss,orsacrificingbasic needs.[87]

ThesituationisespeciallydireinLMICs,wheretheab-senceofUHCandrelianceonout-of-pocketpaymentsplace immensestrainonpatients,oftencausingdelaysoraban-donment of treatments.[87–89]Even in HICs that have no UHC, therisingcostsofmoderntargetedtherapiesandimmuno-therapiessignificantlyincreasefinancialdistressamongpa-tients. A U.S. study reported nearly 20% of patients prescribed oral anticancer agents abandoning or delaying treatment due to highout-of-pocketexpenses.[90]Thesefinancialpressures broadly impact healthcare systems, limiting their capacity for sustainablecancercare.[91–93]Healthsystemsoftenmanage competinghealthpriorities,makinghigh-costcancertreat-mentsparticularlychallenging.

Addressingthisfinancialtoxicityrequiresmulti-levelini-tiatives.[93]Attheindividuallevel,measuressuchasfinancial counseling,patientnavigation,andmedicationassistance programsarecritical.[94]Community-levelinterventionsin-cludeadvocacyandfundraisinginitiativessupportingvul-nerablepatients.[95]Nationally,governmentsshouldintegrate cancer treatments into UHC frameworks, regulate drug prices, and promote cost-effective generics and biosimilars.[93]India's Ayushman Bharat scheme, for instance, expanded cancer care coveragetoreducepatientexpenditures.[96]Globally,multi-lateralagenciesanddonorsshouldchampionpooledpro-curementframeworksandsolidarityfundstoalleviate financialburdensinlow-resourcesettings.TheAccessto Oncology Medicines (ATOM) Coalition, led by the Union for InternationalCancerControl(UICC),exemplifiesthisap-proachbypoolingresourcestonegotiatebetterprocurement termsforessentialcancermedicinesinLMICs.[97,98]

ShortageofSpecialistWorkforce

Thereisawidespreadshortageofoncologyworkforce,with starkcontrastsbetweenHICsandLMICs.[36]Thenumberof cancerdoctorsinAfricannationsrangesfromzeroinsome countriestoonlyafewhundredinothers,meaningmany patientsneverseeanoncologist.[36]Bycontrast,HICshave morespecialistsbutstillfacesignificantshortfallsascancer casesrise.TheUSAanticipatesashortageofover2300on-cologists by 2025,[99]while the UK similarly projects a severe shortageofclinicaloncologists.[100]Ruralandunderserved areas within well-resourced nations also struggle significantly toattractandretainoncologyprofessionals.[101]InSouthAf-rica,theshortageofspecialistsisparticularlysevereinrural publichospitals.[102]
Addressingthischallengerequiresinnovativestrategies suchastask-sharingandtask-shifting,wherenon-specialist healthcareprovidersundertakeoncologyroles.Modelsin-cludetrainingPHCdoctors,clinicalofficers,andnursesin essential oncology skills, enabling them to manage aspects of cancercare.[36]Forinstance,Rwandasuccessfullytrained generalpractitionerstodeliverchemotherapyandmanage common cancers.[69]Nurse-led clinics have expanded oncology access,enablingnursestoindependentlymanagefollow-up care,symptoms,andpalliativecare.[103]InBangladesh,on-cologynursingtrainingwascrucialforestablishingabone marrowtransplantunit.[104]

GapsinPHC–DrivenCancerCare

AlthoughPHCplaysacriticalroleacrossthecancercare continuum—frompreventionandearlydetectiontosurvivor-ship and palliative support—it is often underutilized.[55,56,105,106] Commonchallengesincludeinadequateoncologytraining among PHC providers, limited cancer education in medical and nursing curricula, and insufficient referral systems linking PHC withspecializedoncologyservices.[107]Theseshortcomings

resultindelayeddiagnosis,suboptimalmanagement,and poorer patient outcomes.[55,56]In Botswana, for instance, patients experiencedmediandelaysof160daysfrominitialPHC consultationtospecialistcancertreatment.[108] Thisunderusecreatesgapsincontinuityofcare:once patientsarereferredtospecialistservices,PHCisfrequently sidelined,causingfragmentedfollow-upandlostopportuni-ties for ongoing support.[107]Key underdeveloped areas in PHC includepalliativecareandsurvivorship.InSaudiArabia,a pilotinitiativetrainedfamilyphysiciansinbasicpalliative care,establishingdedicatedclinicdaysatPHCcenters.This approach resulted in earlier interventions, improved symptom control,and90%patientsatisfaction.[109]InCanada,theIN-TEGRATEProjectdemonstratedthatintegratingpalliative careintoprimarycarepracticesincreasedproviders'confi-dence and facilitated earlier advance care planning for cancer patients,addressinggapsincontinuityofcareandcoordi-nation between primary and specialized oncology services.[110] SurvivorshipcareremainsrarelyintegratedintoPHC;even advancedPHCpracticesintheU.S.oftenlackformalsur-vivorshipprogramsduetounclearroles,limitedinformation exchange,andinadequatehealthIT support.[111] Bridgingthesegapsrequiresstrategiesalignedwith AACD'sprinciplesofintegrated,people-centeredcare,in-cluding strengthened bidirectional referral systems, clear post-treatmenttransitionplans,improvedoncologytrainingfor PHC teams, and engaged community support networks.[112,113] HealthsystemswithrobustPHCfoundationsdemonstrate betteroutcomesandlowercosts,evenforcomplexdiseases likecancer.[56,114,115] Ideally,thismeanspatientsreceivetimely,competent evaluationslocally,rapidreferralforspecializedtreatment, andeffectivefollow-upandpalliativesupportathome—a continuum of care anchored in PHC. Such a model aligns with theAACDvision,placingrevitalizedPHCatthecoreof

common,withfewerthan30%specifyingcostsorresource projections,despitefrequentreferencestohealthworkforce strategies. In HICs, common shortcomings include insufficient useofevidence-basedstrategies,limitedalignmentwith globalhealthinitiatives,andlowengagementwithpatient stakeholders, as only 18% of NCCPs involve cancer survivors inplanning.[118]

Policy implementation also varies widely. Thailand provides a notable success story: it achieved UHC in 2002 and integrated cancer care into this system in 2005. By investing in preventive measures, such as tobacco and alcohol control via taxation, and expanding treatment progressively, Thailand aligned its cancer plan with available resources, significantly improving access.[119] Similarly,SouthKorea's10-YearCancerControlPlanin-creasedcancerscreeningratesandimproved5-yearsurvival rates through UHC integration.[119]Japan introduced systematic cost-effectivenessevaluationstomanageoncologycosts,il-lustratingadaptablesustainabilitystrategies.[119]

Nevertheless,significantchallengesremaininmanyre-gions.InNigeria,oncologyservicesareseverelyunder-resourced,withshortagesofspecializedcancerservicesand trained professionals, causing patients to often receive care in general wards under suboptimal conditions.[120]Policy experts in Nigeria emphasize expanding UHC to cover cancer care as crucial.[121]India,despitelongstandingNCCPs,experiences substantialdisparitiesbetweenstatesandhealthcaresectors. Publichospitalsfaceoverwhelmingdemand,resultingin treatmentdelaysaveragingtwomonthspost-diagnosis, whileprivatefacilitiesofferhigh-qualitycareprimarilyto affluentpopulations,highlightinginequitableaccess.[96]Some governmentsfundtheseinitiativesthroughtaxesorcross-subsidies (eg, tobacco/alcohol levies).[122]Kenya, for example, establishedpublic-privatepartnershipstodevelopradiother-apycentersanddiagnosticservices,complementingpublic resources.[123]

equitable, integrated, and people-centered cancer care Robust systems.[2,116] essential for

GovernmentStrategiesandPolicyChallenges

The integration of AACD principles into oncology ultimately dependsoneffectivehealthpolicies.Governmentsplaya decisive role in shaping equitable cancer care through national planning,resourceallocation,healthinsurancedesign,and regulatoryframeworks.Recently,therehasbeengrowing policyrecognitionoftheneedtoprioritizecanceralongside other non-communicable diseases (NCDs). By 2017, the World HealthAssemblyurgedmemberstatestodeveloporupdate NationalCancerControlPlans(NCCPs)aspartoftheNCD agenda,[117,118]resultinginmorethan120countriesim-plementingNCCPs.[118]Theseplanstypicallyaddresspreven-tion,earlydetection,treatment,palliativecare,andresearch.

However,havingaplandoesnotalwaystranslateinto effectiveimplementation.Analysesrevealcriticalgapsin manyNCCPs.InLMICs,inadequatefinancialplanningis

assessment (HTA) guides decisions on the adoption, coverage, andpricingofinterventions.[124]InHICs,agencieslikethe FoodandDrugAdministration(FDA)andEuropeanMedi-cines Agency (EMA) approve new cancer drugs, while entities liketheUK'sNationalInstituteforHealthandCareExcel-lence(NICE)conductHTAstodeterminetherapycoverage, influencingtheavailabilityandpricingofinnovations.[124,125] Often, novel drugs are initially not cost-effective at list price, promptingpayerstodemandpricereductionsorconditional reimbursementuntilsufficientdataemerge.[125,126]Chinaap-plied HTA before adding immunotherapy drugs to its national formulary,achievingpricecutsofnearly80%.[127]Trastuzu-mab's high cost delayed its uptake in LMICs, with fewer than 10%includingitintheirformulariesuntillower-costbio-similarsbecameavailablelater.[128]

To effectively operationalize NCCPs, health systems must strategically prioritize interventions. Table 1 presents practical measuresalongsidefeasibleexamples.

Table1.Practicalmeasuresandexamples.

Measures	Examples
PhasedimplementationofNCCPs HTAinitiatives	Thailand'sincrementalcancerc areexpansion[119] China'sHTA-
Cost-effectivegenericsandbiosimilars	
Public-privateinfrastructurepartnerships	

Licensingandpooledprocurementofbiosimilarsinsub-SaharanAfrica[129]	Kenya'sradiotherapycenters[123]
Cross-subsidiesortaxation	Tobacco/alcoholtaxation[119]

PoliticalandInstitutionalResistance

Someinstitutions,suchasnationalcancerhospitalsorspecialtynetworks,mayresistreformsperceivedtodilutetheir resourcesorauthority.Limitedbudgetsandinfrastructureat thePHClevelfurtherimpedeitsintegrationintocancer care.[130]In practice, these dynamics mean that shifting cancer caretothecommunityrequiresovercomingvestedinterests andreallocatingresourcesagainstthestatusquo.High-level political commitment is often lacking, as policymakers juggle competinghealthprioritiesandmaybeinfluencedbyen-trenched hospital-based lobbies. The result is a slower, uneven uptakeofprogramsincommunitysettings,despiteevidence thatcloser-to-homecarecouldimproveaccessandequity.[130]

Institutionalinertiaandprofessionalsilossimilarlypose challenges to oncology–PHC integration and further advancing AACD principles. Cancer care has traditionally been the domain ofspecialists,sometimesleadingtoa"captureand norelease"model where patients stay under oncology follow-up in hospitals longafteracutetreatment.Thisculturecanfosterresistanceto shared-caremodels.Oncologistsmaydoubtprimarycarepro-viders' oncology expertise or be reluctant to relinquish control of survivorshipcare,whilegeneralpractitionersoftenlackclear role definitions, training, and resources for cancer management.[131,132]Systemicfactorsreinforcethedivide:fee-for-servicepaymentmodelsandhospitalbudgetsrewardin-tensive,specialist-ledinterventions,creatinglittleincentiveto shiftaspectsofcaretocommunityclinics.Fragmentedinfor-mationsystemsexacerbatetheproblem–withoutseamless communication channels, hospital and primary teams struggle to coordinate.[131]Evenwell-resourcedsystemsface"powerdy-namics"issues:tertiarycentersandspecialtyprovidershold financial and decision-making power, and reforms that empower PHCmayencounterpushbackwithininstitutionalhierarchies.

Conclusion

Achieving meaningful progress in global cancer care requires fully embracing the AACD principles of equity, accessibility, prevention, and community participation. By aligningcancer

care.[133]TheAstanaConferencereaffirmedtheenduringim-portance of the AACD,[4]yet further efforts are essential to fully integrate its valuable legacy into contemporary cancer control. To operationalize this vision in global oncology, the following healthpolicyprinciplesshouldbefollowed:

·Ensureequityincancercare:Prioritizeimplementing policies that allocate resources and services in a manner thatprioritizesunderservedpopulations.Thisincludes investingincancerfacilitiesandworkforceinlow-resourceareasandreducingfinancialbarrierstocare, so that the benefits of new treatments and early detection areequitablyrealizedacrossallsegmentsofsociety.·ImproveaccessibilitythroughPHCintegration:Inte-gratebasiconcologyservicesintoPHCnetworksto bring cancer prevention, diagnosis, and treatment closer towherepeoplelive.Thismeansstrengtheninghealth infrastructure and referral systems at the community and primary-carelevel,trainingfrontlineprovidersines-sential oncology skills, and ensuring availability of vital medicines.
·Prioritize prevention and early detection: Make prevention thecornerstoneofglobalcancercontrolbyexpanding programs for vaccination, tobacco control, healthy lifestyle promotion,andenvironmentalhazardreduction.Equally importantisestablishingwidespreadearlydetectionand screeninginitiativesthroughprimarycare.
·Fostercommunityparticipationincancercontrol:Ac-tively involve communities in the planning, im-plementation,andevaluationofcancercareinitiatives.
Thisentailspartneringwithlocalleaders,community health workers, patient advocacy groups, and civil society organizationstoensureprogramsarewidelyaccepted.

Adopting an approach rooted in these principles will guide ourcommunitiesandnationstowardstrongerhealthsystems andbettercanceroutcomes,ultimatelybringingtheworld **closer to the BM quantum computers leverage furniture goal of** “health for all” in the realm ofcancercare.

controlstrategieswiththeseprinciples,policymakersand health leaders can create a more just and effective framework forcancerpreventionandtreatment.Thisapproachcallsfor ensuringthatallpopulations–regardlessofsocioeconomic status or geography – benefit equally from advances in cancer

ORCIDiDs
Javier-DavidBenitez-Fuentes

AlvaroRodriguez-Lescure

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